

[Draft fragment] Test-edge leakage inflates seven years of signed-link prediction benchmarks; a strict protocol and audit restore comparability

[Authors]

1 Introduction

Signed networks — where edges carry a positive (cooperation, trust, agreement) or negative (interference, distrust, disagreement) sign in addition to topological information — have become the canonical benchmark domain for evaluating sign-aware graph neural networks. Five publicly available datasets (Bitcoin-Alpha, Bitcoin-OTC, Slashdot, Epinions, Reddit Hyperlinks) have anchored a seven-year methodological trajectory: SGCN [?], SiGAT [?], SGCL [?], SiGformer [?], SE-SGformer [?], and DADSGNN [?], with peer-reviewed AUROC ceilings rising from ~ 0.85 to ≥ 0.99 on Bitcoin-Alpha and Bitcoin-OTC under a *transductive* evaluation regime.

We show that this trajectory does not reflect a corresponding improvement in sign-prediction capability. The standard transductive protocol leaks test-edge sign information into the model’s feature pipeline at training time, allowing every published method we audited to recover the sign of held-out edges from features that should not have seen it. The leakage is not a bug in any specific implementation — it is a structural property of how signed-graph message-passing is currently set up. We diagnose it with a simple *label-shuffle audit*, define a *strict training-edge-only protocol* that closes the leak, re-evaluate the audited methods under the corrected protocol, and present a compact architecture — Gömb-strict — that remains competitive under the corrected protocol with 30 000–100 000 parameters and a total compute budget of ~ 24 GPU-hours on a single consumer GPU.

The corrected leaderboards differ from the published ones by [Phase B : $\Delta_{\min}$ – $\Delta_{\max}$] percentage points on average across methods and datasets; we recommend the 2018–2025 signed-link-prediction literature be re-evaluated under the strict protocol.

2 Why the transductive protocol leaks

Signed graph neural networks construct per-vertex features by aggregating signed messages from incident edges. In every method we audited, the signed-incidence matrix $\mathbf{H}_{\text{signed}} \in \{-1, 0, +1\}^{|V| \times |E|}$ used in this aggregation is constructed once from the *full* edge list, before the train/test split is applied. Concretely:

$$\mathbf{H}_{\text{signed}}[u, e] = \begin{cases} +1 & \text{if } u \in e \text{ and } \text{sign}(e) = +1, \\ -1 & \text{if } u \in e \text{ and } \text{sign}(e) = -1, \\ 0 & \text{otherwise.} \end{cases} \quad (1)$$

The train/test split is then applied only to the *supervisory signal* — which edges’ signs are masked from the loss — not to the *feature construction*. As a result, when a message-passing

layer computes vertex features as $\mathbf{h}_v = \sigma(\sum_{e \ni v} \mathbf{H}_{\text{signed}}[v, e] \mathbf{W} \mathbf{h}_v^{(e)})$, the sign of every held-out test edge incident to v is read directly from $\mathbf{H}_{\text{signed}}$ and propagated into v 's representation. Backpropagation updates $\mathbf{W}$ using gradients computed from these leaked features. At test time, the trained model predicts the sign of a held-out edge e^* using vertex representations that already contained $\text{sign}(e^*)$ as an input.

The same mechanism applies to all derived signed-graph features: signed Laplacians [?], balance-theoretic triad indicators [?], and structural-equivalence metrics. None of these can be computed without the full signed adjacency, and none of the audited methods rebuilds them per training mini-batch from training-edge signs only.

The fix is straightforward in principle and invasive in practice: during training, every signed-feature artifact must be constructed from training-edge signs alone, with test-edge contributions zeroed out or replaced by a special "unknown" symbol. We call this the *strict training-edge-only protocol* and define it formally in Section ??.

A label-shuffle audit makes the leakage visible without committing to any architectural fix: train under the standard protocol, then randomly permute the test-edge signs (preserving the training-edge signs), and measure test AUROC. If the feature pipeline is clean of test-edge information, shuffled-test AUROC should equal chance (0.5 in expectation). If the feature pipeline leaks, shuffled-test AUROC remains substantially above chance, in proportion to the amount of test-edge sign information the features carry. The audit is a falsifiable, model-agnostic test: any signed-link-prediction method that fails it is, by definition, using information it should not.

References

- [1] [TODO/CW] SGCN — Derr et al., ICDM 2018.
- [2] [TODO/CW] SiGAT — Huang et al., CIKM 2019.
- [3] [TODO/CW] SGCL — Shu et al., AAAI 2022.
- [4] [TODO/CW] SiGformer — author/venue/year.
- [5] [TODO/CW] SE-SGformer — author/venue/year.
- [6] [TODO/CW] DADSGNN — author/venue/year.
- [7] F. Cartwright & D. Harary, "Structural balance: A generalization of Heider's theory," *Psychological Review* 63(5):277–293, 1956.